CAREER: Physics As Architecture: Principled Physics Inductive Biases For Robust, Efficient, And Interpretable Machine Learning

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Overview

Deploying AI in scientific and safety-critical domains — from 3D scene reconstruction to wildfire forecasting — demands models that generalize from limited observations, reduce data ingestion and energy consumption, and produce physically consistent predictions that domain scientists can interpret and trust. Current architectures struggle on all three fronts because physical knowledge, when incorporated at all, is applied only as a soft training penalty on an otherwise unconstrained model, i.e., Physics-Informed Neural Networks (PINNs). These models yield solutions that approximate physical laws statistically rather than satisfying them structurally. This proposal aims to lay the foundations for a paradigm shift: **physics as an architectural inductive bias**, where models are parameterized, by construction, as solution families of the governing equations of the phenomena they represent — going beyond existing approaches that incorporate physics only as a soft training signal. The result is a new class of machine learning models that are compact, interpretable, and physically consistent — not by penalization, but by design. We organize the research into three synergistic thrusts, detailed below.

Intellectual Merit

The proposed research contributes to the foundations of physics-informed machine learning through the three synergistic thrusts. **Thrust 1** will introduce a principled methodology for encoding PDE solution structure as a hard architectural constraint in deep learning architectures, moving beyond dominant soft-regularization methods, such as PINNs. This thrust will be validated through application to state-of-the-art 3D reconstruction, exploiting its established connection to physics. **Thrust 2** will formalize the role of conservation laws and physical symmetries as inductive biases in spatiotemporal multi-agent systems, i.e., modeled as graphs of interacting agents, providing a theoretically grounded path toward data-efficient multi-agent learning. **Thrust 3** will extended the framework developed in Thrusts 1–2 to wildfire propagation modeling. Wildfire propagation serves as the driving application — a high-impact domain where accurate, real-time forecasting remains an open scientific challenge — providing a rigorous testbed for evaluating the generalizability and scalability of physics-as-architecture beyond controlled settings.

Broader Impact

This proposal will catalyze a transformative shift in AI development, impacting a wide range of adjacent areas in the near future. The proposed research agenda will develop a framework for physics-informed AI methods that are more interpretable and robust, directly benefiting industries such as precision agriculture, wildlife management, and disaster response, while requiring significantly less data and computational resources — making AI adoption more accessible and sustainable beyond academia. This research agenda will integrate educational and outreach components for AI literacy, fostered within an **AI Forge Club**. This club is conceived as an interdisciplinary space where graduate and undergraduate students will collaborate to prototype AI applications through annual Build-a-Thons, interact with industry speakers, and participate in mentorship programs to gain research exposure and prepare for future career challenges. The PI has successfully piloted the AI Forge Club since 2025, through Research experience for undergraduate programs at UTA and external sources. Moreover, the club will include **collaborations and recruitment at K12 schools across North Texas and rural underrepresented areas**, serving as a career-preparation STEM educational tool by helping students develop physics-grounded AI rather than simply operate "black-box" models, while inspiring interest in foundational STEM.